Diffraction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain spreading around edges and through openings.

SUCCESS CRITERIA - I can...

1. Compare opening size with wavelength.
2. Distinguish diffraction from refraction.
3. Use a wave picture to justify spreading.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “diffraction” and “wavelength” to explain your model or result. Sentence starters are optional.

diffraction: wave spreading around an edge or through an opening

wavelength: distance between corresponding points on repeated waves

boundary: the place where a medium or constraint changes

01 Warm-up: bring back an earlier idea

A wave travels 9 m in 0.3 s at frequency 50 Hz.
Find speed and wavelength.

Word help: frequency = number of cycles each second

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

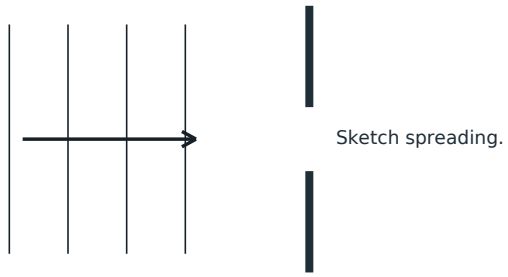
02 Worked example: explain the model

Reference: λ = v/f; compare opening width with λ. Spreading is pronounced when the opening width is comparable to wavelength. Label the opening and wavefronts.

Worked reasoning: A wave with wavelength 1 m meets openings 1 m and 10 m wide. Spreading relative to opening size is more noticeable for the 1 m opening.

Your turn: Draw incoming parallel wavefronts and the emerging patterns for each opening.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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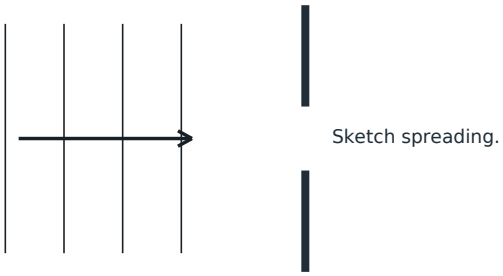
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Sketch wavefronts after a narrow opening about one wavelength wide.
Compare with a much wider opening.

Sentence starter: Both … . They differ in … because … .

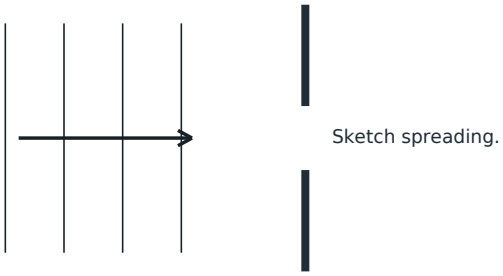


Marzano check - circle: ① ② ③

04 Apply the physics

A 340 Hz sound travels at 340 m/s toward a doorway 0.8 m wide.
Calculate its wavelength and compare wavelength/opening ratios for this sound and visible light of wavelength 5.0×10^{-7} m.
Use wavefront sketches to explain why hearing around a doorway differs from seeing around it.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: ① ② ③

05 Explain your reasoning

A student calls all wave bending “refraction.” Explain how diffraction can happen without crossing into a different-speed medium.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

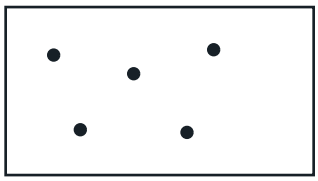
06 Review: connect an earlier idea

For a fixed ideal-gas particle mass, average translational kinetic energy becomes 16 times as large.
Compare rms speed.
Use particle arrows and labels to show the comparison.

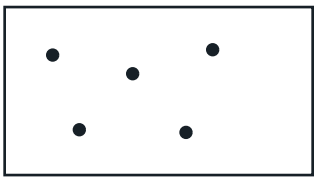
Word help: particle = a small constituent, such as an atom or molecule

Sentence starter: Both … . They differ in … because … .

Sample A



Sample B



Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: ① ② ③

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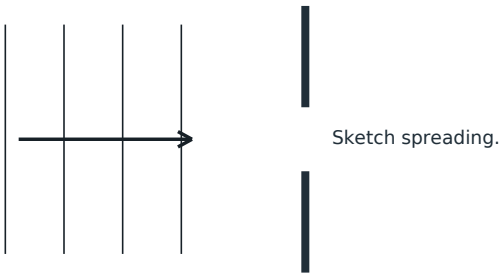
Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

At fixed wave speed, frequency doubles.
Explain how λ and typical diffraction through a fixed opening change.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



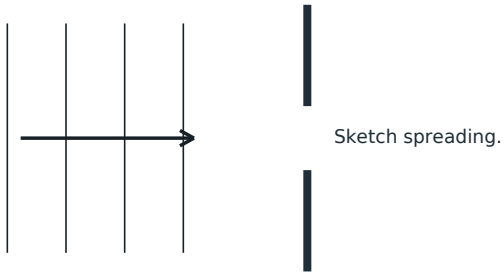
Sketch spreading.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Sound speed is 340 m/s.
Compare wavelengths at 170 and 1700 Hz with a 1 m doorway.
Explain which should diffract more noticeably.

Sentence starter: Both ... . They differ in ... because ... .



Sketch spreading.

Marzano check - circle: 1 2 3